

Global Tuberculosis Burden Analysis (1990–2013)

A WHO Data Analysis Report

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Executive Summary

This report answers a set of core questions about the global burden of tuberculosis (TB), using WHO data covering 219 countries from 1990 to 2013.

The question

- Is the world making real progress against tuberculosis, and is that progress shared everywhere or only in some places?

What we were asked to do

- Clean and prepare the raw WHO dataset.
- Analyze it to answer five groups of questions: overall trends, regional comparison, HIV co-infection, health system performance, and how reliable the estimates are.
- Present the findings visually in an easy to read dashboard.

What we did

- Removed empty or unneeded columns and renamed the rest so they are easy to read.
- Checked the data for errors. Confirmed there were no duplicate entries and that every numeric figure was stored correctly as a number.
- Loaded the cleaned data into PostgreSQL and wrote queries to answer each question directly.
- Built the charts and map shown in this report using Power BI and Tableau.

Issues found in the data

- One column was entirely empty and was removed.
- 65 of the 219 countries have no HIV co-infection estimate for any year. WHO was simply unable to produce this figure for them. We did not fill these gaps with zero, since that would wrongly suggest "no HIV cases" instead of "no data available." These countries were flagged and excluded from HIV related averages.
- A similar, smaller gap exists in case detection rate.
- Outside of these flagged gaps, the dataset was complete and reliable.

What we found

- Global TB incidence, prevalence, and death rates all fell between 1990 and 2013, with the sharpest declines after roughly 2000.
- Progress is uneven. Africa and South East Asia carry by far the highest TB burden, several times higher than Europe or the Americas.
- HIV co-infection is concentrated in Africa. About 27% of TB cases there are HIV-positive, compared to 3% or less in most other regions.
- Deaths among HIV-positive TB patients rose through the 1990s and only began falling around 2002. Deaths among HIV-negative TB patients declined steadily the whole time.
- Countries that are better at finding and diagnosing TB cases tend to have fewer deaths from it. This holds in every region, strongest in South East Asia and the Western Pacific, weakest in the Americas.
- Global case detection rates improved over the period.

What this means

- "Global progress" is real but not evenly shared. Africa and South East Asia still carry the heaviest burden and need to be treated as distinct priorities, not folded into a global average.
- TB and HIV programs were not fully integrated in the 1990s. Closer coordination between the two is where the biggest remaining gains are likely to come from.
- Continued investment in diagnostic capacity is a sound, evidence backed way to reduce deaths, especially outside the Americas where detection and mortality are less closely linked.
- The 65 countries with no HIV estimate are a real blind spot. Any global HIV-TB figure should be read with that gap in mind.

1. Background and Objective

The source data is WHO's Tuberculosis Burden by Country dataset: 5,120 records covering 219 countries across all 6 WHO regions, from 1990 to 2013. For each country and year, it reports TB prevalence, mortality, incidence, HIV co-infection, and case detection rate, each with a low and high bound reflecting WHO's uncertainty.

The analysis covers five groups of questions: how global TB burden has changed over time, how burden compares across regions and countries, the relationship between TB and HIV, whether health system performance is linked to outcomes, and how reliable the underlying estimates are.

2. Data Preparation

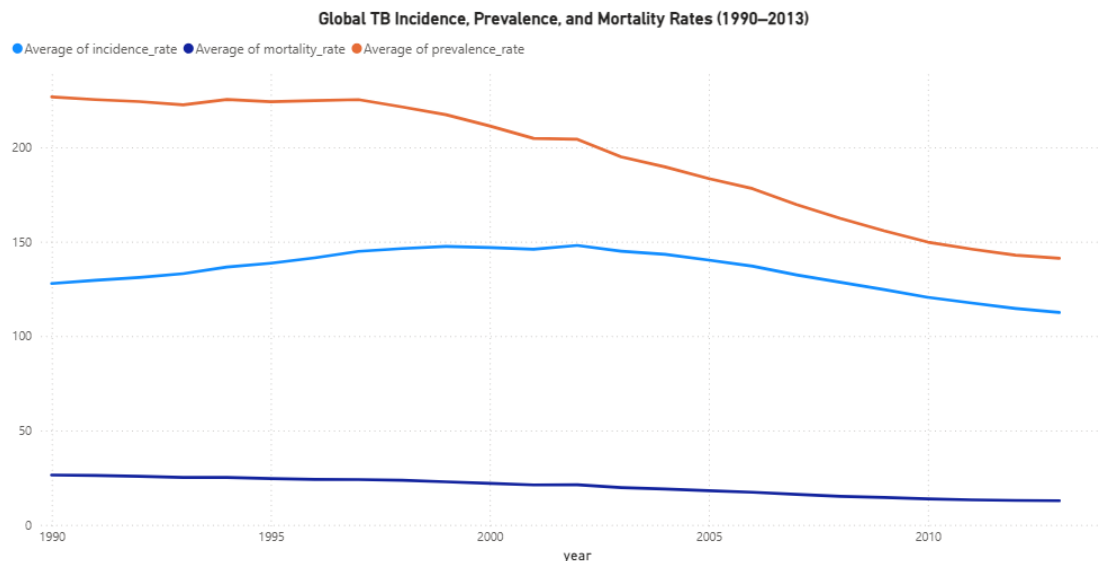
Before analysis, the raw file was cleaned in Python:

- Removed one column that was completely empty, plus a few other columns not needed for this analysis.
- Standardized all column names into a short, consistent format.
- Verified every column held the correct type of data: text fields as text, numeric fields as numbers.
- Checked for duplicate records. None were found.
- Found that 65 of 219 countries have no HIV estimate in any year, while the remaining 154 have one every year. These 65 were flagged and excluded from HIV related averages.

3. Findings by Question

3.1 Has the global TB burden declined since 1990?

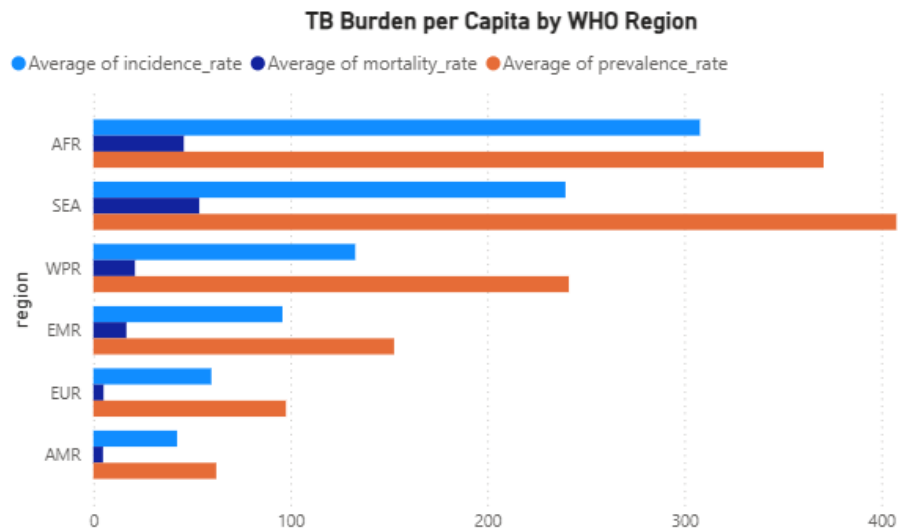
Yes. TB incidence, prevalence, and mortality rates all trended downward from 1990 to 2013, with the clearest declines from the early to mid 2000s onward. Mortality was already comparatively low and continued a slow, steady decline.



Global TB incidence, prevalence, and mortality rates, 1990–2013 (rates per 100,000 population)

3.2 Which regions and countries carry the highest TB burden?

Averaged across 1990 to 2013, Africa (AFR) and South East Asia (SEA) carry by far the largest burden, in both new cases and total existing cases:



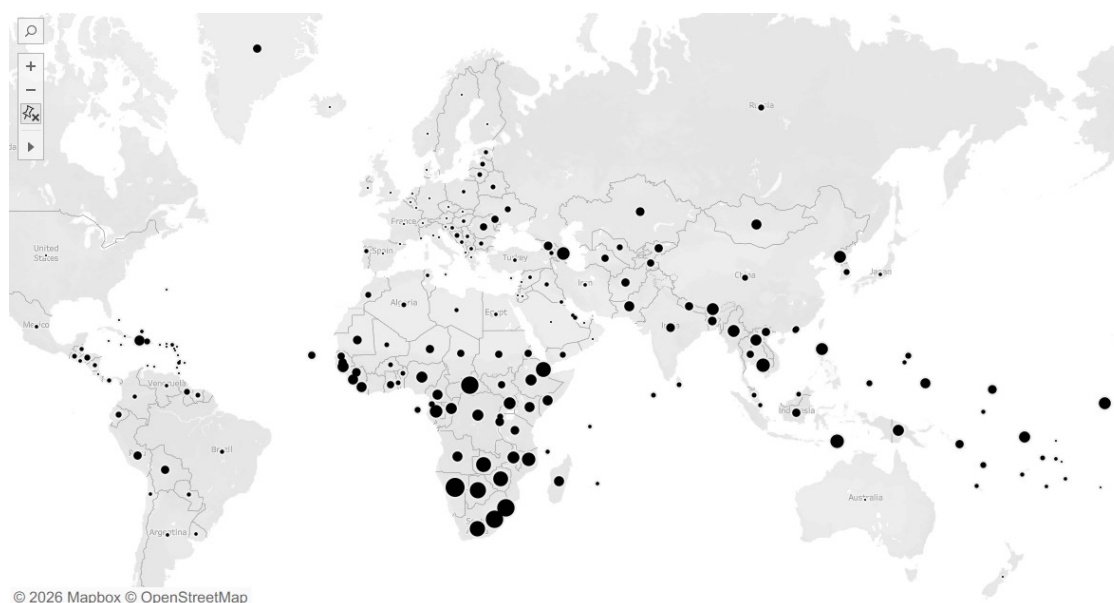
TB burden per capita by WHO region, averaged across 1990–2013

2013 alone, the most recent year, tells the same story, though every region improved compared to its 24 year average:

Region	Avg. Incidence Rate	Avg. Prevalence Rate	Avg. Mortality Rate
AFR	269.5	291.1	30.5
SEA	220.7	310.2	27.9
WPR	108.3	163.1	11.2
EMR	87.9	134.5	14.0
EUR	37.9	50.9	3.4
AMR	30.7	41.7	3.1

Regional TB burden, 2013 only (rates per 100,000 population).

AFR and SEA remain roughly 7 to 9 times more burdened than AMR, the lowest burden region, even after decades of global decline. The map below shows the same pattern by country. Larger, darker markers mean higher incidence, concentrated across Central and Southern Africa and parts of South and South East Asia.



TB incidence rate by country (marker size reflects incidence rate)

3.3 How does HIV co-infection affect the picture, and does it vary by region?

Among the 154 countries with an HIV estimate, the share of TB incidence that is HIV-positive varies widely by region:

Region	Avg. Share of TB Incidence that is HIV-positive	Countries with an Estimate
AFR	26.77%	45
AMR	11.25%	29
SEA	3.14%	10
EUR	2.47%	45
WPR	2.33%	14
EMR	1.67%	11

Average HIV share of TB incidence by region, and how many countries in each region had an estimate.

Africa stands apart. Roughly one in four TB cases there is also an HIV case, far above any other region. It is also the region with the fewest countries missing HIV data (only 2 of 47), so this figure is well supported.

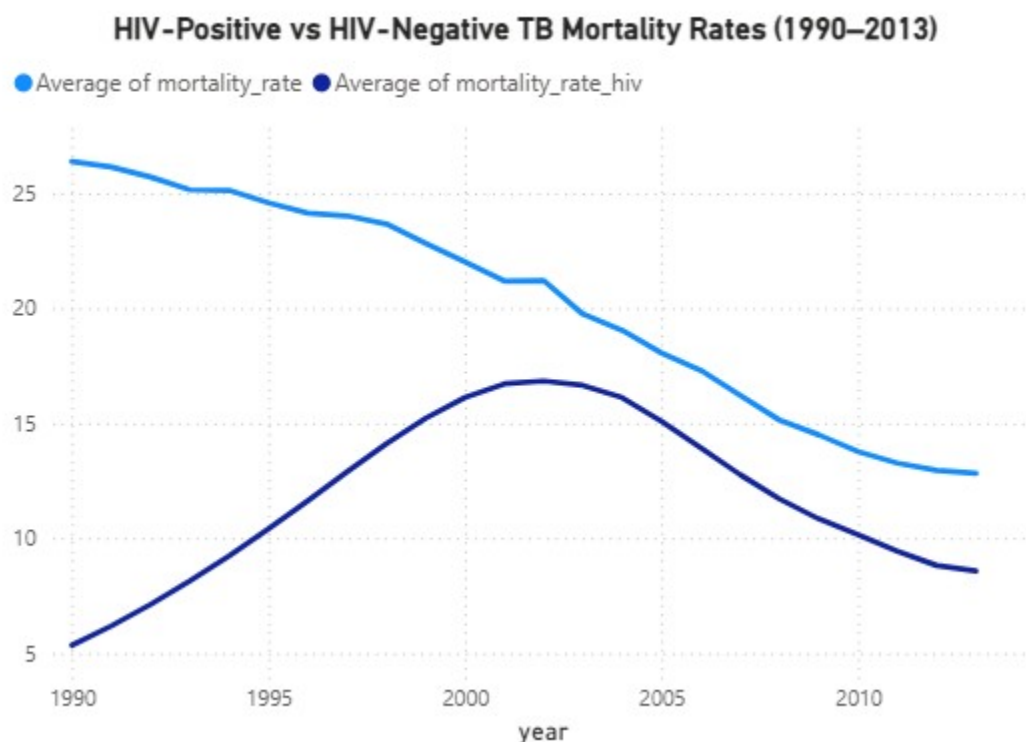
The gaps in HIV reporting are not evenly spread. The Western Pacific (22 countries), the Americas (19), and the Eastern Mediterranean (11) have the most countries with no HIV estimate at all, which limits how confident we can be about the true picture there.

Region	Countries with No HIV Estimate at All
WPR	22
AMR	19
EMR	11

EUR	10
AFR	2
SEA	1

Number of countries per region with no HIV estimate in any year (out of 219 total).

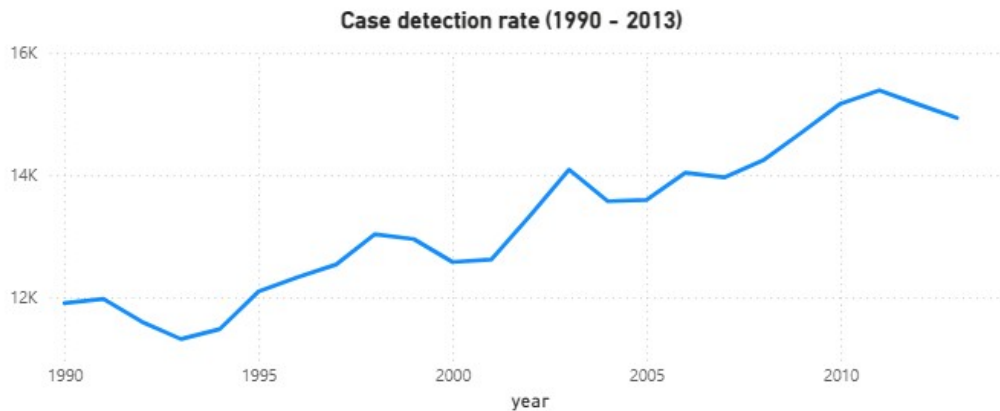
We also compared death rates among HIV-positive and HIV-negative TB patients. HIV-negative TB mortality fell steadily from 1990 onward. HIV-positive TB mortality rose through the 1990s, peaked around 2001 to 2002, and only then began to decline. This group did not share in the earlier progress the rest of the TB affected population saw; their improvement started roughly a decade later.



HIV-positive vs. HIV-negative TB mortality rates, 1990–2013

3.4 Is better case detection linked to fewer deaths?

Case detection rate measures how well health systems find and diagnose TB cases that actually exist. Globally, it improved over the period, particularly from the early 2000s onward.



Global TB case detection rate, 1990–2013

There is also a clear statistical relationship between detection and mortality: higher detection rates line up with lower death rates (correlation of -0.573 overall, on a scale from -1 for a perfect inverse relationship to +1 for a perfect direct one). This supports the idea that finding TB cases earlier gives patients a real chance to be treated before the disease becomes fatal.

This relationship holds in every region, but its strength varies:

Region	Correlation (Detection vs. Mortality)	Number of Records
SEA	-0.682	243
WPR	-0.652	765
EUR	-0.554	1,213
AFR	-0.553	1,035
EMR	-0.436	491
AMR	-0.149	924

Correlation between case detection rate and mortality rate, by region.

The relationship is strongest in South East Asia and the Western Pacific, and clearly weakest in the Americas. In the Americas, factors other than detection, such as treatment access or drug resistance, may play a larger role in outcomes.

3.5 How reliable are these estimates, and is that improving?

Every WHO estimate comes with a low and high bound reflecting the range of plausible true values. The width of that range (high bound minus low bound) shows how confident WHO is in a given figure. A narrow range means solid underlying data; a wide range means the number relies more on modeling. Whether this range narrowed over time speaks to whether global TB surveillance genuinely improved, separate from whether the disease itself declined.

4. What This Means for Decision-Making

- Treat "global progress against TB" as real but unevenly shared. Africa and South East Asia still carry a disproportionate share of the burden and should be treated as distinct priorities.
- HIV-TB co-infection is a large, specific challenge concentrated in Africa, and it was the slowest improving part of the overall picture. Integrating TB and HIV testing and treatment is likely where the biggest remaining gains are.
- Investing in case detection is a sound, evidence backed way to reduce deaths, given the consistent link between detection and mortality across every region. Where that link is weaker, particularly the Americas, it is worth checking whether the bottleneck sits elsewhere in care, such as access to treatment once a case is found.
- The 65 countries with no HIV data, and the broader gaps in case detection reporting, are themselves worth flagging. Any global figure built on this data should be read with that gap in mind.